Chapter 1

Primitive Notions

To build a deductive theory of geometry we must first fix the very language in which the theory will be expressed. Like any mathematical discipline, geometry begins with a collection of *primitive* concepts - terms that we do not define in terms of anything simpler but whose mutual relationships are governed by a system of axioms. From these axioms, we will derive theorems regarding the behavior and properties of these primative notions. For our axiomatic system we adopt three primitive notions: one kind of object, and two relations among those objects. Together they form a single structure that captures the intuitive idea of a flat plane, and will later be used to prove represents a flat plane given our chosen axioms.

The first primitive notion is the **point**. Points are the individuals of our theory; they are the entities that occupy locations in the plane without themselves having parts or extension.

On this set we impose two primitive relations. The first is a *ternary* relation $\mathcal{B}$ ("betweenness"), and the second is a *quaternary* relation $\cong$ ("congruence"). These three ingredients - the point-set and the two relations - are the sole raw material from which the whole theory will be constructed.

The entire structure, consisting of the domain $\mathbf{P}$ equipped with the relations $\mathcal{B}$ and $\cong$, is what we call the **plane** and denote by $\mathcal{P}$.

1.1 Plane

Before dissecting the individual notions, it is useful to definine the whole apparatus formallys.

Definition 1 (Plane)**.** *A plane is a structure*

$$\mathcal{P} := \langle \mathbf{P}, \mathcal{B}, \cong \rangle,$$

where $\mathbf{P}$ *is an arbitrary set,* $\mathcal{B}$ *is a ternary relation on* $\mathbf{P}$*, and* $\cong$ *is a quaternary relation on* $\mathbf{P}$*.*

Thus, saying "let $\mathcal{P}$ be a plane" amounts to specifying a set of points together with a particular interpretation of betweenness and congruence satisfying the axioms we shall gradually introduce. The definition deliberately uses the neutral term "a plane" rather than "the Euclidean plane" because, until we make further choices, the axioms we give will be true in a broad class of geometries.

1.2 Points

The elements of the domain $\mathbf{P}$ are termed *points*. We denote points by upper-case Latin letters: $A, B, C, \ldots$. All points are gathered into a set $\mathbf{P}$, called the *point-set*, defined by

$$\mathbf{P} := \{ P \mid P \text{ is a point} \}.$$

Within the formal system, a point is an object whose only properties are those granted to it by the axioms governing $\mathcal{B}$ and $\cong$. There is no further internal structure: a point is defined not by decomposition into simpler parts, but solely by its membership in the set $\mathbf{P}$.

Axiom 1 (Existence of Distinct Points)**.** $\exists A, B \in \mathbf{P}, A \neq B$.

This axiom guards against the trivial one-point model satisfying every later axiom.

Informal characterization. Although the formal theory treats points as primitives, it is helpful to keep in mind the intuitive picture that guides our choice of axioms. A point can be thought of as a unique, indivisible position in the plane. This corresponds to the classical Euclidean characterization of a point as "that which has no part." Informally, a point is regarded as having no magnitude - no length, width, or depth - and as being zero-dimensional. However, this physical description is only a heuristic. Within the deductive framework, the formal behaviour of points is entirely determined by the relations in which they stand, and the intuitive characterization carries no logical weight in the proofs that follow. We draw on the geometric imagery only to motivate the axioms and to make the abstract relations vivid.

1.3 Betweenness

Betweenness is a ternary relation on the set of all points, denoted by $\mathcal{B} \subseteq \mathbf{P}^3$. Given points A, B, and C, we write $\mathcal{B}(A, B, C)$ and read it as "B lies between A and C".

Unlike a set $\{A, B, C\}$, where the order of elements is immaterial, betweenness is defined over the cartesian product $\mathbf{P} \times \mathbf{P} \times \mathbf{P}$. Thus it depends on the order of the arguments. Formally,

$$\mathcal{B} \subseteq \mathbf{P}^3 := \{(A, B, C) \mid A, B, C \in \mathbf{P}\}.$$

The expression $\mathcal{B}(A, B, C)$ is simply the assertion that $(A, B, C) \in \mathcal{B}$.

In Euclidean geometry a point B is between A and C precisely when the three points lie on a line and B is met while traveling from A to C along that line.Our axioms capture this idea without invoking lines or order directly.

We now x five axioms governing the behavior of $\mathcal{B}$.

1.3.1 Axioms of Betweenness

Axiom 2 (Symmetry of Betweenness)**.**

$$\forall A, B, C \in \mathbf{P}, \mathcal{B}(A, B, C) \implies \mathcal{B}(C, B, A).$$

Betweenness is symmetric with respect to the outer points: if B lies between A and C, then it also lies between C and A. This reflects the fact that the segment AC is the same as the segment CA. Left-to-right implication is used over a biconditional due to redundancy, as the right-to-left implication directly follows from this form of the axiom.

Axiom 3 (Order)**.**

$$\forall A, B, X \in \mathbf{P},\ \mathcal{B}(A, B, X) \implies \big(\neg\mathcal{B}(B, A, X)\ \wedge\ \neg\mathcal{B}(A, X, B)\big).$$

If B is between A and X, then A cannot be between B and X, nor can X be between A and B. In other words, a point can serve as a "middle" in only one way: the triple (A, B, X) determines a unique order along the line. Together with the Identity axiom, this implies that

$\mathcal{B}(A, A, A)$ is impossible (otherwise, taking $A = B = X$, we would have both $\mathcal{B}(A, A, A)$ and its negation). Thus the relation is *non-reflexive* in the strict sense: no point lies between itself and itself.

Axiom 4 (Identity of Betweenness)**.**

$$\forall A, B \in \mathbf{P}, \mathcal{B}(A, B, A) \implies A = B.$$

This axiom says that if a point appears as the "middle" of a triple that begins and ends with the same point, then the two outer points must coincide. In particular, it forbids the possibility that a distinct point B B lies between A and A; such a configuration would contradict the intuition that betweenness requires a genuine interval. Notice that the converse, $A = B \implies \mathcal{B}(A, A, A)$, is *not* assumed; indeed the next axiom will exclude it.

Axiom 5 (Density)**.**

$$\forall A, C \in \mathbf{P}, A \neq C \implies \exists B \in \mathbf{P} \colon \mathcal{B}(A, B, C).$$

Whenever two points are distinct, there exists a point strictly between them. This captures the idea that segments are dense—between any two points one can always find a third.

The plane must be infinite. Thus, we introduce an axiom that will later allow us to guarenteed the existence of infinite points along a line.

Axiom 6 (Extensibility)**.** *Given two points A and B, there exists a point C such that* $\mathcal{B}(A, B, C)$

$$\forall A, B \in \mathbf{P}, A \neq B \implies \exists C \in \mathbf{P} \colon \mathcal{B}(A, B, C).$$

Axiom 7 (Transitivity of Betweenness)**.** $\mathcal{B}(A, B, C) \wedge \mathcal{B}(B, C, D) \wedge B \neq C \implies \mathcal{B}(A, B, D)$.

If B lies between A and C, and C lies between B and D (with $B \neq C$), then B lies between A and D. This axiom allows us to chain betweenness relations along a line.

These six axioms are mutually independent: no one can be derived from the others (as can be shown by standard model-theoretic arguments). Together with the congruence axioms to be introduced later, they will form the basis of our geometric theory.

1.3.2 Immediate Consequences

As is, the extensibility axiom allows only for extensibility to the right. To prove this consisely for both directions we introduce a lemma demostrating extensibility on the left side:

Theorem 1 (Left-Sided Extensibility)**.** *For any two points A and B, there exists C such that $\mathcal{B}(C, B, A)$.*

Proof. From $A \neq B$, the extensibility axiom 6 implies $\mathcal{B}(A, B, C)$. We then apply the symmetry of betweenness 2 to obtain $\mathcal{B}(C, B, A)$, thus arriving at the desired result. □

With this, we can now prove extensibility on both sides.

Theorem 2 (Dual-Sided Extensibility)**.** *Given any two points B and C, there exists points A and D such that $[ABCD]$.*

Proof. Let $B, C \in \mathbf{P}$ be two points not equal to each other. From the extensibility axiom. there exists a point D such that $[BCD]$. Similarily, from left-sided extensibility, there exists a point A such that $[ABC]$. These results are sufficent to demostrate that $[ABCD]$, thus proving the theorem. □

Theorem 3. $\mathcal{B}(A,B,C) \iff \mathcal{B}(C,B,A)$ *for all* $A, B, C \in \mathbf{P}$.

Proof. The forward direction is the axiom; the reverse follows by applying the axiom with A and C exchanged. □

Given two distinct points, there is always a point beyond the second, so that the second lies between the first and the new point. This guarantees that lines extend indefinitely.

These five axioms are mutually independent: no one can be derived from the others (as can be shown by standard model-theoretic arguments). Together with the congruence axioms to be introduced later, they will form the basis of our geometric theory.

Theorem 4 (Irreflexivity of Betweenness)**.** *No point lies between it and itself:*

$$\forall A \in \mathbf{P},\ \neg\mathcal{B}(A,A,A).$$

Proof. Assume $\mathcal{B}(A,A,A)$. By the order axiom, this implies $\neg\mathcal{B}(A,A,A)$ and $\neg\mathcal{B}(A,A,A)$, which contradicts our assumption. Hence, $\neg\mathcal{B}(A,A,A)$. □

Theorem 5 (Distinct Endpoints)**.** *Betweenness forces distinct outer points:*

$$\mathcal{B}(A,B,C) \implies A \neq C.$$

Proof. Suppose $\mathcal{B}(A,B,C)$ and $A = C$. Then, we have $\mathcal{B}(A,B,A)$, which by the identity of betweenness implies $A = B$. Substituting back gives $\mathcal{B}(A,A,A)$, which contradicts the irreflexivity of betweenness. Therefore $A \neq C$. □

Theorem 6 (Irreflexivity of Betweenness for Distinct Points)**.**

$$A \neq B \implies \neg\mathcal{B}(A,A,B) \wedge \neg\mathcal{B}(A,B,B).$$

Proof. If $\mathcal{B}(A,A,B)$, then by the order of betweeness, □

Theorem 7 (Strict Betweenness)**.** *If* $\mathcal{B}(A,B,C)$*, then* $A \neq B$ *and* $B \neq C$*.*

Proof. Suppose $\mathcal{B}(A,B,C)$ and $A = B$. Then, $\mathcal{B}(A,A,C)$, which implies $\neg\mathcal{B}(A,A,C)$ by the order of betweenness. However, this is a contradiction. Thus, $A \neq B$. Similarly, if we assume $B = C$, then we have $\mathcal{B}(A,B,B) \implies \neg\mathcal{B}(A,B,B)$, leading to a contridiction. Thus, $B \neq C$. □

Theorem 8 (Contrapositive Ordering)**.** *For all* $A, B, X \in \mathbf{P}$*,*

$$\mathcal{B}(B,A,X) \implies \neg\mathcal{B}(A,B,X) \quad \textit{and} \quad \mathcal{B}(A,X,B) \implies \neg\mathcal{B}(A,B,X).$$

Proof. The Order axiom gives the two implications

$$\mathcal{B}(A,B,X) \implies \neg\mathcal{B}(B,A,X) \quad \text{and} \quad \mathcal{B}(A,B,X) \implies \neg\mathcal{B}(A,X,B).$$

Taking the logical contrapositive of each implication(Iwhich is logically equivalent to the original), yields precisely the two stated results. □

Theorem 9 (Symmetry Extended)**.** *Given* $\mathcal{B}(A,B,C)$ *none of the following hold:* $\mathcal{B}(B,A,C)$*,* $\mathcal{B}(A,C,B)$*,* $\mathcal{B}(B,C,A)$*,* $\mathcal{B}(C,A,B)$*.*

These theorems complete our baseline analysis of how the betweenness relation $\mathcal{B}$ operates locally on individual triples of points. However, proving that the plane $\mathcal{P}$ is structurally infinite—both in its internal density and its outward extension—requires the formal definitions of segments and lines. We defer these proofs to chapter blank, where the infinite nature of the geometric line can be properly established.

1.4 Congruence

Congruence is a quaternary relation on the set of all points denoted by $\cong \subseteq \mathbf{P}^4$. Rather than mapping individual points to one another, this relation establishes a geometric equivalence between two pairs of points. We write $\cong (A,B,C,D)$, to assert that the segment from point A to point B is congruent to the segment from point C to point D. Formally:

$$\cong (A,B,C,D) \iff (A,B,C,D) \in \cong .$$

To reflect this conceptually, we bundle the point pairs (A,B) and (C,D) into the typographical units AB and CD and introduce the infix symbol $\cong$:

$$AB \cong CD \overset{\text{def}}{\iff} \cong (A,B,C,D)$$

If the relation holds for the pairs (A,B) and (C,D), we write $(A,B) \cong (C,D)$, which is interpreted as the "distance" from A to B being equal to the distance from C to D. The notation for the points will be reused to define segments. This choice was intentional, as the later informal usage of congruence upon segments is identical in nature to what is presented below.

We now impose three axioms governing the behavior of $\cong$.

1.4.1 Axioms of Congruence

Axiom 8 (Reversal of Congruence)**.**

$$\forall A,B \in \mathbf{P},\ AB \cong BA.$$

Axiom 9 (Transitivity of Congruence)**.**

$$\forall A,B,C,D,E,F \in \mathbf{P},\ \big(AB \cong CD \land AB \cong EF\big) \implies CD \cong EF.$$

Transitivity (Axiom 9) is the formal way of saying that two segments congruent to the same segment are congruent to each other.

With only Reversal and Transitivity, however, the axioms do not distinguish between genuine segments (whose endpoints are distinct) and degenerate ones (whose endpoints coincide). Model in which every point pairs is congruent to one another would strictly satifsy the above. In this universe, distinct points exist (Axiom 1), but they are not separated. Such a “zero-distance” universe reduces the congruence relation to the universal relation and makes the geometry collapse: all points are at the same distance from one another, and no meaningful notion of length can be recovered.

To exclude this degenerate model we introduce the following.

Axiom 10 (Identity of Congruence)**.**

$$\forall A, B, C \in \mathbf{P},\ AB \cong CC \implies A = B.$$

In words: if a segment is congruent to a degenerate segment, then its endpoints must coincide. Equivalently, a non-degenerate segment ($A \neq B$) can never be congruent to a degenerate one (CC). Intuitively, this means the "distance" between distinct points is never zero, — a requirement that, in a metric geometry, would be imposed by the condition $d(A, B) = 0 \iff A = B$.

Note that the axiom does not guarenteed the existence of distinct points — that guarantee was already given by the domain axiom Axiom 1. However, these axioms together guarantee that the plane contains at least two points *and* that distinct points are geometrically distinct.

Once distinct, genuinely separated points exist, **??** populates the segment between them with infinitely many intermediate points, ensuring that the line is not a discrete set of isolated locations.

1.4.2 Congruence as an Equivalence Relation

An equivalence relation is characterized by three properties: reflexivity, symmetry, and transitivity. The axioms adopted here for $\cong$ assert neither reflexivity nor the general symmetry and transitivity required of an equivalence relation directly; Transitivity, in particular, is stated in a shared-first-argument form rather than the familiar chain form. The remainder of this subsection shows that Reversal and Transitivity nonetheless force Reflexivity, Reflexivity together with Transitivity forces Symmetry, and Symmetry together with Transitivity forces the familiar chain form of transitivity, so nothing further needs to be assumed.

Theorem 10 (Reflexivity of Congruence)**.**

$$\forall A, B \in \mathbf{P},\ AB \cong AB.$$

Proof. From reversal (Axiom 8), we have $AB \cong BA$. Through replacement of variables we also have $BA \cong AB$. We now instantiate transivity (Axiom 9) with $AB \cong BA$ used twice:

$$\big(AB \cong BA \text{ and } AB \cong BA\big) \implies BA \cong BA.$$

Next, we apply transivity again as follows:

$$\big(BA \cong AB \text{ and } BA \cong AB\big) \implies AB \cong AB,$$

which completes the proof. □

With reflexivity established, the other expected properties follow rapidly.

Theorem 11 (Symmetry of Congruence)**.** *If $AB \cong CD$, then $CD \cong AB$.*

Proof. We are given $AB \cong CD$. From the reflexivity axiom we also have $AB \cong AB$. Applying these two congruences to the transitivity axiom gives the antecedent

$$AB \cong CD \wedge AB \cong AB,$$

which implies $CD \cong AB$. □

Theorem 12 (Full-Transitivity of Congruence)**.** *If $AB \cong CD$ and $CD \cong EF$, then $AB \cong EF$.*

Proof. By hypothesis, $AB \cong CD$. Applying Symmetry (Theorem 11) to this gives

$$CD \cong AB.$$

Together with the second hypothesis $CD \cong EF$, we obtain the conjunction

$$CD \cong AB \ \wedge \ CD \cong EF.$$

This is precisely an instance of the Transitivity axiom (Axiom 9), instantiated with CD in the shared first-argument position of both conjuncts. The axiom therefore yields

$$AB \cong EF,$$

as required. □

With these results, $\cong$ is established as an equivalence relation on ordered pairs of points: reflexive and symmetric by theorem, and transitive in the familiar chain sense by Full-Transitivity. Consequently, $\cong$ partitions the set of all ordered point pairs into disjoint equivalence classes, each consisting of all segments congruent to one another — the geometric counterpart of segments of equal length. This is the extent of what an equivalence relation alone provides: it identifies which segments0 share a length, but supplies no order between distinct classes and no arithmetic operation upon them. Such structure requires the interaction of $\cong$ with $\mathcal{B}$ and is developed in a later section.

1.4.3 Endpoint-Reversal Symmetries

The equivalence-relation properties established above govern the behavior of $\cong$ across distinct pairs of points; they say nothing about the effect of reordering the two endpoints *within* a single pair. This subsection establishes that $\cong$ is in fact invariant under such reordering, in every combination.

Theorem 13 (Left-Reversal of Congruence)**.** *If $AB \cong CD$, then $BA \cong CD$.*

Proof. We are given $AB \cong CD$. From the reversal axiom, we also have $AB \cong BA$. Applying transitivity to the conjunction

$$AB \cong BA \wedge AB \cong CD$$

yields $BA \cong CD$. □

Theorem 14 (Right-Reversal of Congruence)**.** *If $AB \cong CD$, then $AB \cong DC$.*

Proof. We are given $AB \cong CD$. From the reversal axiom applied to the pair (C, D), we have $CD \cong DC$. By symmetry (Theorem 11) applied to the hypothesis, we also have $CD \cong AB$. Applying transitivity to the conjunction

$$CD \cong AB \wedge CD \cong DC,$$

yields $AB \cong DC$. □

Theorem 15 (Full-Reversal of Congruence)**.** *If $AB \cong CD$, then $BA \cong DC$.*

Proof. We are given $AB \cong CD$. From the reversal axiom, we know that $AB \cong BA$. By Right-Reversal (Theorem 14) applied to the hypothesis, we also have $AB \cong DC$. Applying transitivity to

$$AB \cong BA \wedge AB \cong DC,$$

yields $BA \cong DC$. □

The previous three theorems can be packaged into a single convenient equivalence, establishing that segment congruence is entirely indifferent to the order of the endpointsof either segment.

Theorem 16 (Endpoint-Swap Invariance)**.** *For any points A, B, C, D,*

$$AB \cong CD \iff BA \cong DC \iff AB \cong DC \iff BA \cong CD.$$

Proof. The forward implications have all been proved:

- $AB \cong CD \Rightarrow BA \cong DC$ (Full-Reversal)
- $BA \cong DC \Rightarrow AB \cong DC$ (apply Left-Reversal to the pair (B, A))
- $AB \cong DC \Rightarrow BA \cong CD$ (Right-Reversal followed by Left-Reversal)
- $BA \cong CD \Rightarrow AB \cong CD$ (Left-Reversal on (B, A))

The reverse directions follow by the same theorems applied in the opposite order, completing the circle of equivalences. □

1.5 Interaction of Betweenness and Congruence

Axiom 11 (Segment Construction)**.**

$$\forall Q, X, A, B \in \mathbf{P},\ Q \neq X \implies \exists Y \in \mathbf{P} \colon \mathcal{B}(Q, X, Y) \wedge XY \cong AB.$$